SYSCCA 2.0

Geospa - Installation Guide

VERSION 1.0.0

06/05/2026

1. PENDAHULUAN 4

2. PRASYARAT UMUM PENGETAHUAN 4

3. SPESIFIKASI STACK TEKNOLOGI 4

4. PERSIAPAN SISTEM 4

5. OPENSSL 4

5.1. Instalasi 4

5.2. Verifikasi Instalasis 4

6. GIT 5

6.1. Instalasi 5

6.2. Verifikasi Instalasi 5

7. DOCKER 29.X & COMPOSE 5.X 5

7.1. Instalasi 5

7.1.1. Tambahkan GPG Key Resmi 5

7.1.2. Daftarkan Repositori 5

7.1.3. Instalasi 5

7.2. Aktivasi Service 5

7.3. Verifikasi Instalasi 5

7.3.1. Cek Versi 5

7.3.2. Cek Service 6

8. NGINX 6

8.1. Instalasi 6

8.2. Aktivasi Service 6

8.3. Verifikasi Instalasi 6
8.3.1. Cek Versi 6
8.3.2. Cek Service 6

8.4. Optimasi Konfigurasi Nginx 8
8.4.1. Backup Konfigurasi Default 8
8.4.2. Buat SSL DH Param 8
8.4.3. Buat Konfigurasi Nginx Baru 8
8.4.4. Buat Konfigurasi proxy_param Baru 11
8.4.5. Verifikasi Konfigurasi 11
8.4.6. Terapkan Konfigurasi Baru 11

9. FIREWALL 12

9.1. Konfigurasi Firewall 12
9.1.1. Izinkan SSH (WAJIB PERTAMA, agar tidak terkunci dari server!) 12
9.1.2. Izinkan Port Web 12
9.1.3. Izinkan Port Aplikasi Geospa 12

9.2. Aktivasi Firewall 12

9.3. Verifikasi Firewall 12

10. CLONE REPOSITORY 13

10.1. Buat SSH Key 13

10.2. Daftar Alias SSH 14

10.3. Ambil Public Key 14

10.4. Konfigurasi GitHub Deploy Key 14

10.5. Uji Koneksi 15

10.6. Clone Repository via SSH 16

11. SETUP GEOSPA 16

11.1. Buat Paired Keys 16
11.1.1. Buat Direktori Keys & Logs 16
11.1.2. Buat Private & Public Keys 16
11.1.3. Verifikasi 16

11.2. Konfigurasi SSL 17
11.2.1. Buat selfsigned SSL untuk Geospa 17
11.2.2. Buat Skrip auto-renewal SSL Geospa 18

11.3. Konfigurasi environment (.env) **20**
11.3.1. Salin .env.example 20
11.3.2. Konfigurasi .env Fase Pertama 20
11.3.3. Dapatkan Client ID 20
11.3.4. Konfigurasi .env Fase Kedua 21

12. JALANKAN DEPLOYMENT **22**

12.1. Build & Up Aplikasi Geospa **22**

12.2. Publish Geospa ke Internet (Publik) **22**
12.2.1. Salin Konfigurasi Routing Nginx Geospa ke Nginx Host 22
12.2.2. Aktivasi Konfigurasi Nginx Geospa 22
12.2.3. Verifikasi Konfigurasi 22
12.2.4. Terapkan Konfigurasi Baru 22

12.3. Verifikasi Service Geospa **23**

13. MAINTENANCE **23**

13.1. Pengecekan & Pembaruan Manual SSL **23**

13.2. Pengecekan Status Service **23**
13.2.1. Cek Service Nginx 23
13.2.2. Cek Service Docker 23
13.2.3. Cek Service Geospa 23

13.3. Monitoring Service Geospa **23**

13.4. Monitoring Logs Geospa **23**

14. TROUBLESHOOTING **24**

14.1. DNS Hijacking & Kegagalan Resolusi Nama **24**

14.2. Akses Ditolak pada Docker (Permission Denied) **24**

14.3. APT Mengembalikan Status 404 Not Found (Repository Usang) **25**

15. END OF DOCUMENT **25**

1. Pendahuluan

Dokumen ini dibuat untuk memberikan panduan komprehensif mengenai bagaimana mempersiapkan, menguji, dan melakukan instalasi serta deployment aplikasi mendawai Geospatial.

2. Prasyarat Umum Pengetahuan

Untuk dapat mengikuti panduan instalasi dan setup sistem Asterisk, Issabel, serta layanan aplikasi geospasial ini dengan baik, Anda diwajibkan harus familier dan mengerti tentang hal-hal berikut:

- Linux OS
- Sistem jaringan
- Docker
- Git

3. Spesifikasi Stack Teknologi

- **Sistem Operasi:** Ubuntu 24.04 LTS (Noble Numbat)
- **Runtime:** Docker 29.x
- **Database:** PostgreSQL 16
- **Database Extension:** PostGIS 3, pg_trgm
- **Web Server:** Nginx (Reverse Proxy)
- **Process Manager:** Docker Compose 5.x
- **Aplikasi:** Geospa
- **API Key**: Google Maps API(Maps JavaScript API; Places API; Geocoding API; Maps Static API; Street View Static API;)

4. Persiapan Sistem

Lakukan update repositori dan upgrade package sistem ke versi terbaru dan instal package bantuan.

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y ca-certificates curl gnupg2 lsb-release ubuntu-keyring
```

5. OpenSSL

5. 1. Instalasi

```
sudo apt install -y openssl
```

5. 2. Verifikasi Instalasis

```
openssl version
```

6. Git

6.1. Instalasi

```
sudo apt install -y git
```

6.2. Verifikasi Instalasi

```
git --version
```

7. Docker 29.x & Compose 5.x

7.1. Instalasi

7.1.1. Tambahkan GPG Key Resmi

```
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg
```

7.1.2. Daftarkan Repository

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https://download.docker.com/linux/ubuntu $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sudo apt update
```

7.1.3. Instalasi

```
sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
sudo usermod -aG docker $USER
```

(PENTING: Anda harus logout dan login kembali agar perubahan ini berlaku)

```
exit
```

7.2. Aktivasi Service

```
sudo systemctl enable docker   # Aktivasi docker
sudo systemctl start docker    # Jalankan docker
```

7.3. Verifikasi Instalasi

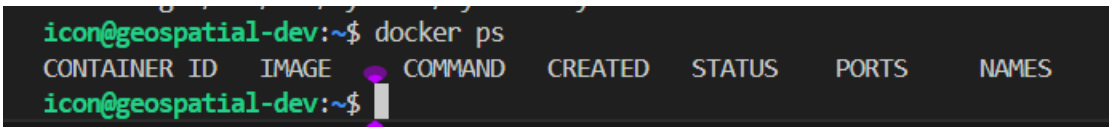
7.3.1. Cek Versi

```
docker -version# Versi terbaru
```

```
docker compose version # Versi terbaru
```

7.3.2. Cek Service

```
docker ps
```



Gambar 7.1. Instalasi Docker & Docker Compose

8. Nginx

8.1. Instalasi

8.1.1. Tambahkan GPG Key Resmi

```
curl -fsSL https://nginx.org/keys/nginx_signing.key | gpg --dearmor | sudo tee
/usr/share/keyrings/nginx-archive-keyring.gpg >/dev/null
```

8.1.2. Daftarkan Repository

```
echo "deb [signed-by=/usr/share/keyrings/nginx-archive-keyring.gpg]
https://nginx.org/packages/mainline/ubuntu $(lsb_release -cs) nginx" | sudo tee
/etc/apt/sources.list.d/nginx.list
echo -e "Package: *\nPin: origin nginx.org\nPin-Priority: 900\n" | sudo tee
/etc/apt/preferences.d/99nginx
sudo apt update
```

8.1.3. Instalasi

```
sudo apt install nginx -y
```

8.2. Aktivasi Service

```
sudo systemctl enable nginx
sudo systemctl start nginx
```

8.3. Verifikasi Instalasi

8.3.1. Cek Versi

```
nginx -v
```

8.3.2. Cek Service

```
systemctl status nginx
```

```
icon@geospatial-dev:~$ nginx -v
nginx version: nginx/1.24.0 (Ubuntu)
icon@geospatial-dev:~$ systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-04-08 20:57:55 WIB; 3min 4s ago
       Docs: man:nginx(8)
   Main PID: 1461852 (nginx)
      Tasks: 5 (limit: 4556)
     Memory: 3.7M (peak: 8.2M)
        CPU: 46ms
     CGroup: /system.slice/nginx.service
             ├─1461852 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─1461854 "nginx: worker process"
             ├─1461855 "nginx: worker process"
             ├─1461856 "nginx: worker process"
             └─1461857 "nginx: worker process"

Apr 08 20:57:55 geospatial-dev systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Apr 08 20:57:55 geospatial-dev systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
```

Gambar 8.1. Instalasi Nginx

8.4. Instalasi Modul Zstd

8.4.1. Persiapan

```
sudo apt update
sudo apt install -y git build-essential libpcre3 libpcre3-dev zlib1g zlib1g-dev libssl-dev
libzstd-dev
```

8.4.2. Mengambil Source Code

```
cd /tmp
git clone --depth 1 https://github.com/tokers/zstd-nginx-module.git
NGINX_VER=$(nginx -v 2>&1 | cut -d '/' -f 2 | cut -d ' ' -f 1)
wget https://nginx.org/download/nginx-$NGINX_VER.tar.gz
tar -xvzf nginx-$NGINX_VER.tar.gz
cd nginx-$NGINX_VER
```

8.4.3. Compile Modul

```
CONFARGS=$(nginx -V 2>&1 | sed -n -e 's/^.*arguments: //p')
eval ./configure $CONFARGS --add-dynamic-module=/tmp/zstd-nginx-module
make modules
```

8.4.4. Instalasi

```
sudo mkdir -p /etc/nginx/modules-enabled/
sudo cp objs/ngx_http_zstd_filter_module.so /usr/lib/nginx/modules/
sudo cp objs/ngx_http_zstd_static_module.so /usr/lib/nginx/modules/
sudo rm -rf /tmp/zstd-nginx-module /tmp/nginx-*
# Membuat file konfigurasi loader
echo "load_module modules/ngx_http_zstd_filter_module.so;" | sudo tee /etc/nginx/modules-
enabled/50-mod-zstd.conf
```

```
echo "load_module modules/ngx_http_zstd_static_module.so;" | sudo tee -a /etc/nginx/modules-enabled/50-mod-zstd.conf
sudo systemctl restart nginx
```

8.5. Optimasi Log Rotate Nginx

8.5.1. Edit Konfigurasi

```
sudo tee /etc/logrotate.d/nginx <<EOF
/var/log/nginx/*.log {
        daily
        maxsize 100M
        missingok
        rotate 14
        compress
        delaycompress
        notifempty
        create 640 nginx adm
        sharedscripts
        postrotate
                if [ -f /run/nginx.pid ]; then
                        kill -USR1 \`cat /run/nginx.pid\`
                fi
        endscript
}
EOF
```

8.5.2. Verifikasi Konfigurasi

```
sudo logrotate -d /etc/logrotate.d/nginx
```

8.6. Optimasi Konfigurasi Nginx

8.6.1. Backup Konfigurasi Default

```
sudo cp /etc/nginx/nginx.conf /etc/nginx/nginx.conf.bak
sudo cp /etc/nginx/proxy_params /etc/nginx/proxy_params.bak
```

8.6.2. Buat SSL DH Param

```
sudo openssl dhparam -out /etc/nginx/dhparam.pem 2048
```

8.6.3. Buat Konfigurasi Nginx Baru

```
sudo tee /etc/nginx/nginx.conf <<'EOF'
user nginx;
worker_processes auto;
worker_cpu_affinity auto;
worker_rlimit_nofile 200000;
pid /run/nginx.pid;
include /etc/nginx/modules-enabled/*.conf;

events {
```

```
    worker_connections 16384;
    multi_accept on;
    use epoll;
}

http {
    # ─── Security Hardening ──────────────────────────────────────────
    server_tokens off;
    charset utf-8;

    # ─── MIME & Optimization ─────────────────────────────────────────
    include       /etc/nginx/mime.types;
    default_type  application/octet-stream;

    # ─── TCP / Network Optimization ──────────────────────────────────
    sendfile on;
    tcp_nopush on;
    tcp_nodelay on;
    quic_gso on;

    # ─── Websocket Proxy Variables ───────────────────────────────────
    map $http_upgrade $connection_upgrade_base {
        default upgrade;
        ""      "";
    }

    map $http3 $connection_upgrade {
        "h3"    "";
        default $connection_upgrade_base;
    }

    map $http3 $active_upgrade {
        "h3"    "";
        default $http_upgrade;
    }

    map $http3 $active_host {
        "h3"    $host;
        default $http_host;
    }

    # ─── Buffer Optimization ─────────────────────────────────────────
    client_body_buffer_size   32K;
    client_header_buffer_size  8k;
    client_max_body_size      50M;
    large_client_header_buffers 16 4k;

    # ─── Timeouts & Keepalive ────────────────────────────────────────
    client_body_timeout   12;
    client_header_timeout 12;
    keepalive_timeout     65;
    keepalive_requests  10000;
    send_timeout          10;

    # ─── SSL Hardening (Mozilla Intermediate) ────────────────────────
    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_prefer_server_ciphers on;
    ssl_session_cache   shared:SSL:10m;
```

```
    ssl_session_timeout 1d;
    ssl_session_tickets off;
    ssl_dhparam /etc/nginx/dhparam.pem;

    # ─── Gzip Compression (Fallback)──────────────────────────────
    gzip on;
    gzip_vary on;
    gzip_proxied any;
    gzip_comp_level 2;
    gzip_min_length 1024;
    gzip_buffers 32 4k;
    gzip_types
        text/plain
        text/css
        text/xml
        text/csv
        text/javascript
        text/x-component
        application/json
        application/javascript
        application/x-javascript
        application/xml
        application/xhtml+xml
        application/rss+xml
        application/atom+xml
        image/svg+xml;

    # ─── Zstd Compression ────────────────────────────────────────
    zstd on;
    zstd_comp_level 4;
    zstd_min_length 1024;
    zstd_buffers 32 4k;
    zstd_types
        text/plain
        text/css
        text/xml
        text/csv
        text/javascript
        text/x-component
        application/json
        application/javascript
        application/x-javascript
        application/xml
        application/xhtml+xml
        application/rss+xml
        application/atom+xml
        image/svg+xml;

    # ─── Logging ─────────────────────────────────────────────────────────────────
    # log_format  main  '$remote_addr - $remote_user [$time_local] "$request" '
    #                   '$status $body_bytes_sent "$http_referer" '
    #                   '"$http_user_agent" "$http_x_forwarded_for"';

    # access_log /var/log/nginx/access.log main buffer=64k flush=1s;
    access_log off;
    error_log /var/log/nginx/error.log warn;

    # ─── Includes ────────────────────────────────────────────────────────────────
    include /etc/nginx/conf.d/*.conf;
```

```
    include /etc/nginx/sites-enabled/*;
}
EOF
```

8.6.4. Buat Konfigurasi proxy_param Baru

```
sudo tee /etc/nginx/proxy_params <<'EOF'
proxy_set_header Host               $active_host;
proxy_set_header X-Real-IP          $remote_addr;
proxy_set_header X-Forwarded-For    $proxy_add_x_forwarded_for;
proxy_set_header X-Forwarded-Proto $scheme;
proxy_set_header X-Forwarded-Host   $host;
proxy_set_header X-Forwarded-Port   $server_port;
proxy_set_header Accept-Encoding    $http_accept_encoding;
proxy_set_header Forwarded "for=$remote_addr;proto=$scheme;host=$host";
proxy_http_version 1.1;
proxy_set_header Upgrade            $active_upgrade;
proxy_set_header Connection         $connection_upgrade;
proxy_buffer_size 8k;
proxy_buffers 16 4k;
proxy_busy_buffers_size 16k;
proxy_max_temp_file_size    0;
proxy_temp_file_write_size 8k;
EOF
```

8.6.5. Buat Konfigurasi security_headers

```
sudo mkdir -p /etc/nginx/snippets
```

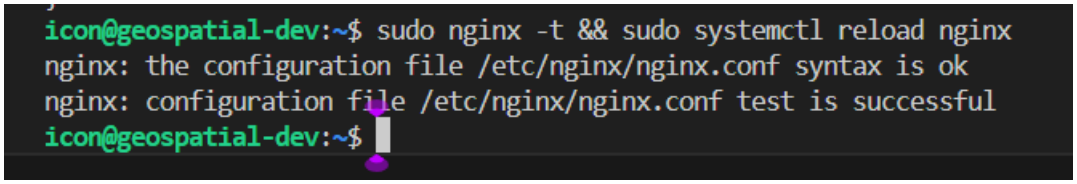
```
sudo tee /etc/nginx/snippets/security_headers.inc <<'EOF'
add_header X-Content-Type-Options "nosniff" always;
add_header X-XSS-Protection "0" always;
add_header Referrer-Policy "strict-origin-when-cross-origin" always;
EOF
```

8.6.6. Verifikasi Konfigurasi

```
sudo nginx -t
```

8.6.7. Terapkan Konfigurasi Baru

```
sudo systemctl reload nginx
```



Gambar 8.2. Tes & Terapkan Konfigurasi Nginx

9. Firewall

9.1. Konfigurasi Firewall

9.1.1. Izinkan SSH (WAJIB PERTAMA, agar tidak terkunci dari server!)

```
sudo ufw allow OpenSSH
```

9.1.2. Izinkan Port Web

```
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
```

9.1.3. Izinkan Port Aplikasi Geospa

```
sudo ufw allow 4170/tcp
sudo ufw allow 4170/udp
sudo ufw allow 3000/tcp
sudo ufw allow 3000/udp
```

9.2. Aktivasi Firewall

```
sudo ufw enable
```

Input 'y' saat diminta konfirmasi.

9.3. Verifikasi Firewall

```
sudo ufw status
```

```
icon@geospatial-dev:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
OpenSSH                    ALLOW       Anywhere
80/tcp                     ALLOW       Anywhere
443/tcp                    ALLOW       Anywhere
4170/tcp                   ALLOW       Anywhere
3000/tcp                   ALLOW       Anywhere
OpenSSH (v6)               ALLOW       Anywhere (v6)
80/tcp (v6)                ALLOW       Anywhere (v6)
443/tcp (v6)               ALLOW       Anywhere (v6)
4170/tcp (v6)              ALLOW       Anywhere (v6)
3000/tcp (v6)              ALLOW       Anywhere (v6)

icon@geospatial-dev:~$
```

Gambar 9.1. Konfigurasi Firewall

10. Clone Repository

10. 1. Buat SSH Key

Buat SSH key khusus untuk repository Geospa dengan format perintah berikut:

```
# Mengandung Formating '< >'(Harap diubah)!!!
ssh-keygen -t ed25519 -C "<name> key" -f ~/.ssh/id_ed25519_<name> -N ""
```

Jika diminta passphrase: tekan Enter (tanpa passphrase untuk server)

```
icon@geospatial-dev:~$ ssh-keygen -t ed25519 -C "geomap-deploy-key" -f ~/.ssh/id_ed25519_geomap -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/icon/.ssh/id_ed25519_geomap
Your public key has been saved in /home/icon/.ssh/id_ed25519_geomap.pub
The key fingerprint is:
SHA256:Lz1pcHldmUghc3nW2J87S1mkaWJPJzVjghN0po2Ho+8 geomap-deploy-key
The key's randomart image is:
+--[ED25519 256]--+
|          .=o*oo.|
|           o%oo*B|
|           =.+=B*|
|          ..=.*o+|
|        S.o..=.o+|
|         =.o  .= |
|        . *.  . o|
|         o..   . |
|           E     |
+----[SHA256]-----+
icon@geospatial-dev:~$
```

Gambar 10.1. Buat SSH Key

10.2. Daftar Alias SSH

Daftarkan SSH key yang dibuat sebelumnya dengan format perintah berikut:

```
# Buat file config jika belum ada
touch ~/.ssh/config
# Mengandung Formating '< >' (Harap diubah)!!!
cat << 'EOF' >> ~/.ssh/config
Host github.com-<name>
  HostName github.com
  User git
  IdentityFile ~/.ssh/id_ed25519_<name>
  IdentitiesOnly yes
EOF
```

10.3. Ambil Public Key

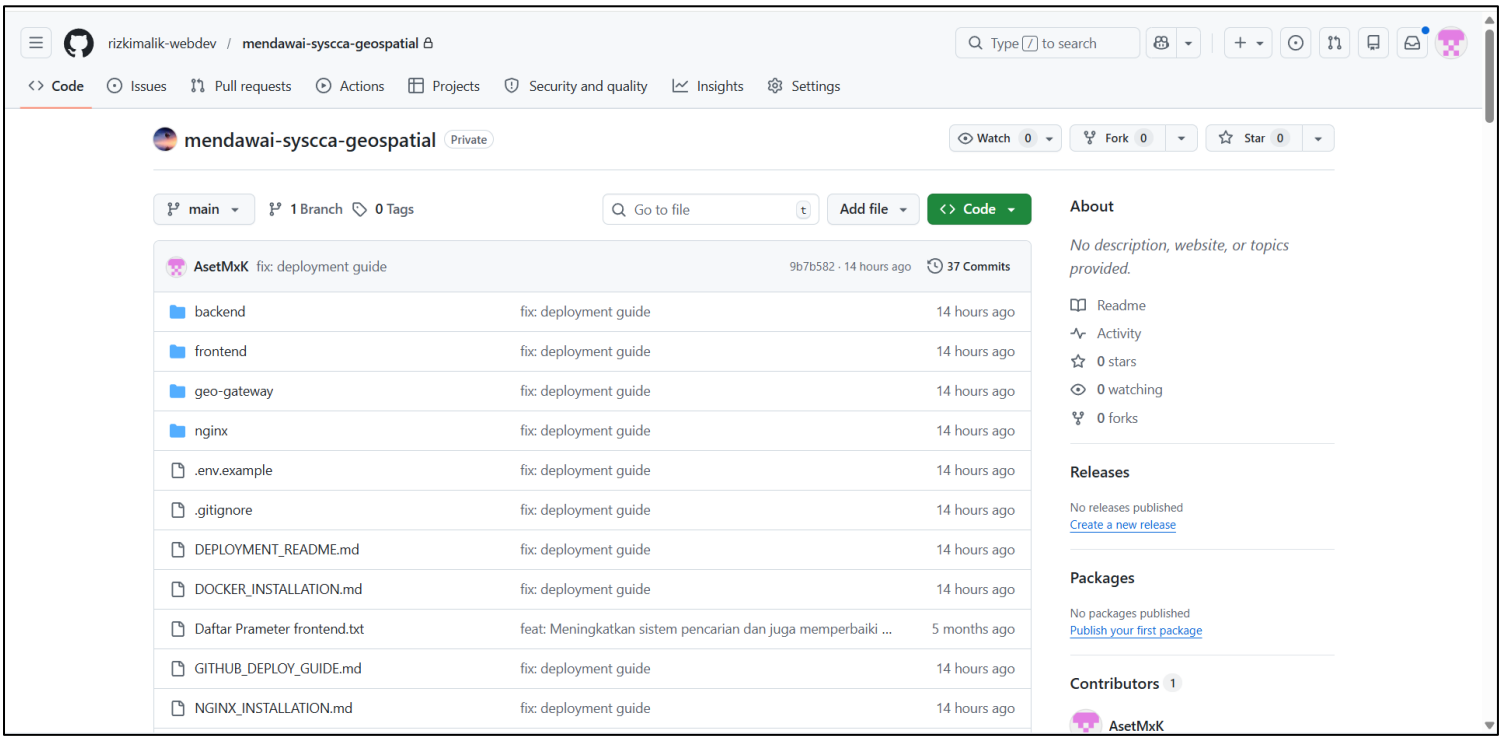
Ambil public key dari SSH key yang dibuat sebelumnya dengan format perintah:

```
# Mengandung Formating '< >' (Harap diubah)!!!
cat ~/.ssh/id_ed25519_<name>.pub
```

Salin semua outputnya (dimulai dari ssh-ed25519) dan simpan.

10.4. Konfigurasi GitHub Deploy Key

1. Buka repository → **Settings** → **Security and quality** → **Deploy Keys**

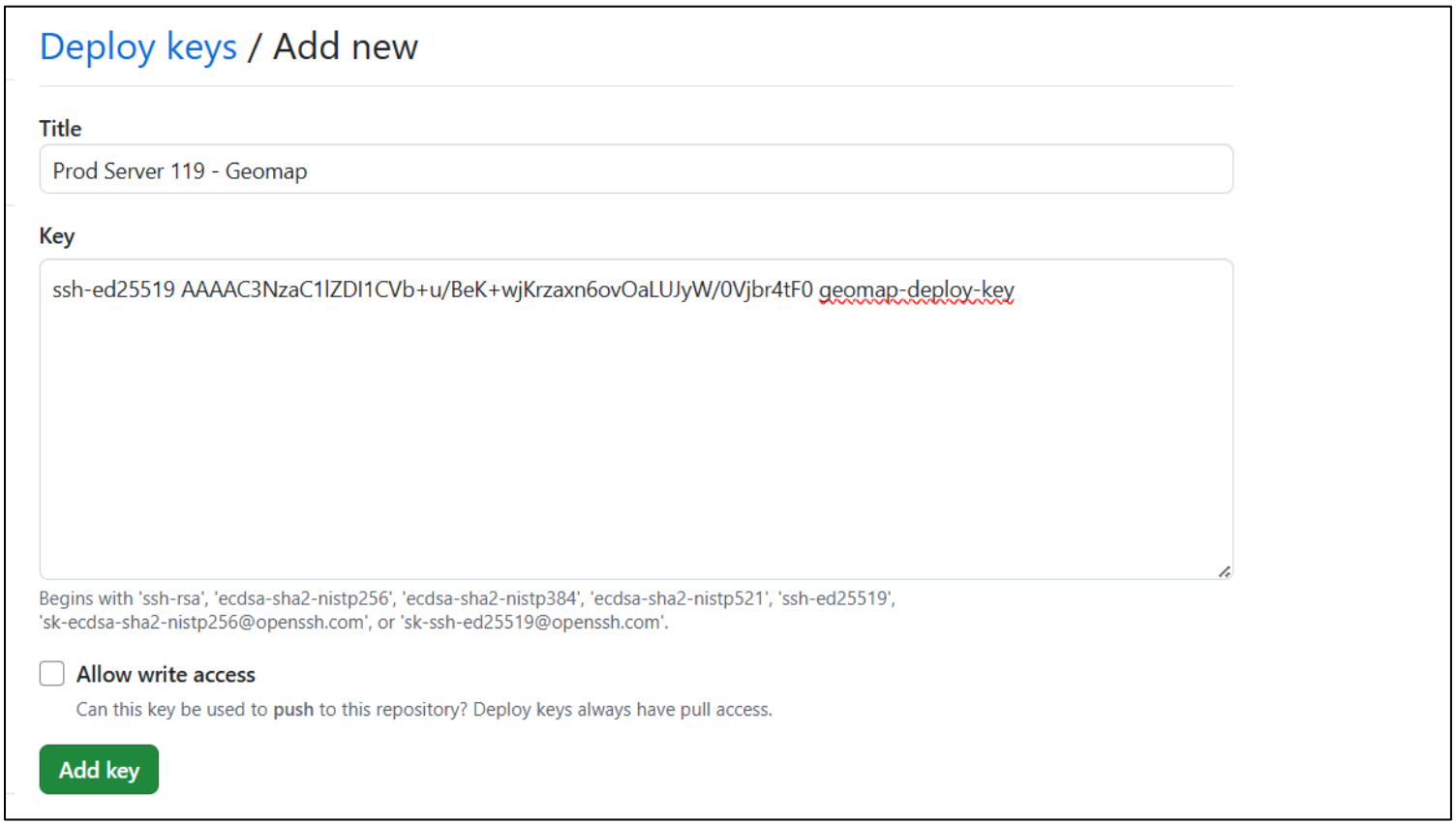


Gambar 10.2. Buka Repository



Gambar 10.3. Masuk ke Settings → Security and quality → Deploy Keys

2. Klik **Add deploy key**



Gambar 10.4. Buat Deploy Key Baru

3. Input **Title**
4. Input **Key** dengan Public key yang sudah disalin pada langkah Ambil Public Key
5. Jangan centang **Allow write key** (Biarkan Read-only demi keamanan server)
6. Klik **Add key**

10.5. Uji Koneksi

Uji koneksi repository dengan format perintah berikut:

```
# Mengandung Formating '< >' (Harap diubah)!!!
ssh -T git@github.com-<name>
```

```
icon@geospatial-dev:~$ ssh -T git@github.com-geomap
Hi AsetMxK/Geomap! You've successfully authenticated, but GitHub does not provide shell access.
icon@geospatial-dev:~$
```

Gambar 10.5. Uji Koneksi Repository dengan Server

10.6. Clone Repository via SSH

Perhatikan output Uji Koneksi. " Hi AsetMxK/Geomap ..." adalah <username>/<repository> anda. Kita butuh nilai itu untuk melakukan cloning dengan format perintah:

```
cd /home/icon
# Mengandung Formating '< >'(Harap diubah)!!!
git clone git@github.com-<name>:<username>/<repository>.git <dir_name>
```

```
icon@geospatial-dev:~$ git clone git@github.com-geomap:AsetMxK/Geomap.git geomap
Cloning into 'geomap'...
remote: Enumerating objects: 669, done.
remote: Counting objects: 100% (669/669), done.
remote: Compressing objects: 100% (439/439), done.
remote: Total 669 (delta 410), reused 463 (delta 210), pack-reused 0 (from 0)
Receiving objects: 100% (669/669), 869.87 KiB | 800.00 KiB/s, done.
Resolving deltas: 100% (410/410), done.
icon@geospatial-dev:~$
```

Gambar 10.6. Cloning Repository via SSH

11. Setup Geospa

11.1. Buat Paired Keys

11.1.1. Buat Direktori Keys & Logs

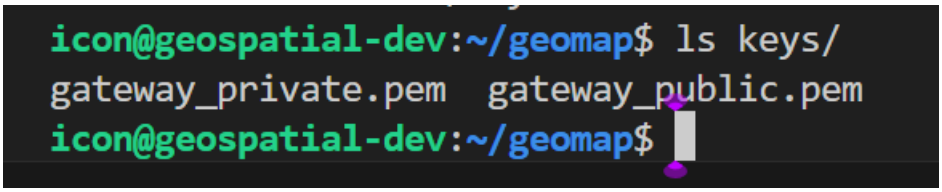
```
# Mengandung Formating '< >'(Harap diubah)!!!
cd /home/icon/<dir_name>
mkdir -p keys/ssl logs
```

11.1.2. Buat Private & Public Keys

```
# Buat private key
openssl genpkey -algorithm RSA -out keys/gateway_private.pem -pkeyopt rsa_keygen_bits:2048
# Buat public key
openssl rsa -pubout -in keys/gateway_private.pem -out keys/gateway_public.pem
```

11.1.3. Verifikasi

```
ls keys/
```



Gambar 11.1. Private & Public Keys

11.2. Konfigurasi SSL

11.2.1. Buat selfsigned SSL untuk Geospa

Buat skrip berikut:

```
cat << 'EOF' > generate_ssl.sh
#!/bin/bash
# 🌊 Optimized Self-Signed SSL Generator with SAN support

set -e

# 1. Deteksi IP Otomatis atau Gunakan Argumen
# Jika menjalankan ./generate_ssl.sh 1.2.3.4 maka akan pakai IP tersebut.
# Jika tidak, akan mendeteksi IP utama server.
DEFAULT_IP=$(hostname -I | awk '{print $1}')
SERVER_IP=${1:-$DEFAULT_IP}

KEY_DIR="./keys/ssl"
CONFIG_FILE="./keys/ssl/openssl.cnf"
mkdir -p $KEY_DIR

echo "🔍 Target IP: $SERVER_IP"

# 2. Buat file konfigurasi OpenSSL sementara
cat << EOC > $CONFIG_FILE
[req]
default_bits       = 2048
distinguished_name = req_distinguished_name
x509_extensions    = v3_req
prompt             = no

[req_distinguished_name]
C  = ID
ST = DKI Jakarta
L  = Jakarta
O  = Kiero Teknologi
OU = IT Department
CN = $SERVER_IP

[v3_req]
basicConstraints = critical, CA:TRUE
keyUsage = critical, digitalSignature, nonRepudiation, keyEncipherment, keyCertSign
extendedKeyUsage = serverAuth
subjectAltName = @alt_names

[alt_names]
IP.1 = $SERVER_IP
```

```
IP.2 = 127.0.0.1
DNS.1 = localhost
EOC

echo "🔐 Generating SSL Certificate (Valid for 10 years)..."

if openssl req -x509 -nodes -days 3650 -newkey rsa:2048 \
    -keyout "$KEY_DIR/nginx.key" \
    -out "$KEY_DIR/nginx.crt" \
    -config $CONFIG_FILE; then

# Hapus file konf sementara
    rm $CONFIG_FILE

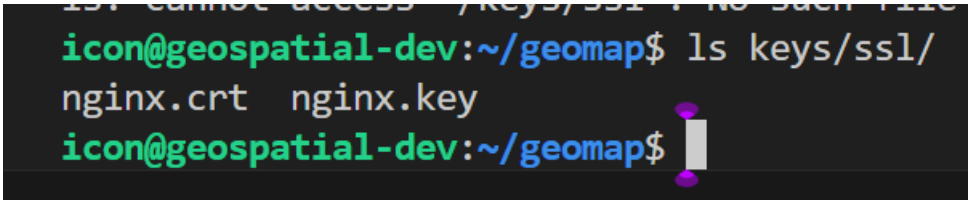
    echo "☑ Success! Certificate generated for $SERVER_IP"
    echo "📍 Location: $KEY_DIR/nginx.crt"
else
    echo "❌ Error: Gagal membuat sertifikat."
    exit 1
fi
EOF
```

Jalankan skrip:

```
chmod +x generate_ssl.sh
./generate_ssl.sh
```

Verifikasi SSL:

```
ls keys/ssl/
```



Gambar 11.2. Self-signed SSL Geospa

Salin SSL ke direktori host nginx:

```
sudo cp keys/ssl/nginx.crt /etc/ssl/certs/geospa-selfsigned.crt
sudo cp keys/ssl/nginx.key /etc/ssl/private/geospa-selfsigned.key
```

11.2.2. Buat Skrip auto-renewal SSL Geospa

Buat skrip berikut:

```
# Mengandung Formating '< >'(Harap diubah)!!!
cat << 'EOF' > renew_ssl.sh
#!/bin/bash
# 🔄 SSL Auto-Renewal Script for Self-Signed Certs
```

```
set -e

PROJECT_DIR="/home/icon/<dir_name>"
CERT_FILE="$PROJECT_DIR/keys/ssl/nginx.crt"
DAYS_BEFORE_EXPIRY=30

echo "🗓️ Checking SSL certificate expiry..."

# 1. Cek apakah file sertifikat ada
if [ ! -f "$CERT_FILE" ]; then
    echo "❌ Certificate not found at $CERT_FILE. Generating new one..."
    cd $PROJECT_DIR && ./generate_ssl.sh
else
    # 2. Cek sisa umur sertifikat dalam detik
    # openssl x509 -checkend akan return 0 jika cert masih valid sampai N detik ke depan
    EXPIRY_SECONDS=$((DAYS_BEFORE_EXPIRY * 24 * 3600))

    if openssl x509 -checkend $EXPIRY_SECONDS -noout -in "$CERT_FILE"; then
        echo "☑️ Certificate is still valid for more than $DAYS_BEFORE_EXPIRY days. No
renewal needed."
        exit 0
    else
        echo "⚠️  Certificate is expiring soon (less than $DAYS_BEFORE_EXPIRY days).
Renewing..."
        cd $PROJECT_DIR && ./generate_ssl.sh
    fi
fi

# 3. Salin ke direktori sistem Nginx (Requires sudo)
echo "📂 Copying new certificates to system directory..."
sudo cp "$PROJECT_DIR/keys/ssl/nginx.crt" /etc/ssl/certs/geospa-selfsigned.crt
sudo cp "$PROJECT_DIR/keys/ssl/nginx.key" /etc/ssl/private/geospa-selfsigned.key

# 4. Reload Nginx
echo "🚀 Reloading Nginx..."
if sudo systemctl reload nginx; then
    echo "------------------------------------------------"
    echo "✨ Renewal process completed successfully."
else
    echo "❌ Error: Nginx failed to reload."
    exit 1
fi
EOF
```

Setup auto-renewal dengan cron job:

```
chmod +x renew_ssl.sh
crontab -e
```

Tambahkan baris ini di paling bawah:

```
# Mengandung Formating '< >'(Harap diubah)!!!
# menjalankan pengecekan setiap tanggal 1 jam 00:00
0 0 1 * * /home/icon/<dir_name>/renew_ssl.sh >> /home/icon/<dir_name>/logs/ssl_renew.log 2>&1
```

11.3. Konfigurasi environment (.env)

11.3.1. Salin .env.example

```
# Mengandung Formating '< >'(Harap diubah)!!!
cd /home/icon/<dir_name>
cp .env.example .env
```

11.3.2. Konfigurasi .env Fase Pertama

Edit dengan editor:

```
nano .env
```

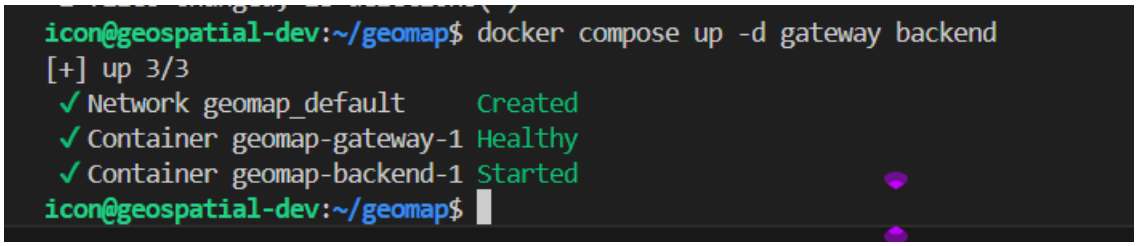
Isi nilai variabel-variabel berikut:

Variabel	Deskripsi	Contoh
BACKEND_DATABASE_URL	URL Koneksi Database postgreSQL	postgresql://user123:password1!@11.15.141.123:5432/backend_db?schema=public&connection_limit=16
GATEWAY_DATABASE_URL	URL Koneksi Database postgreSQL	postgresql://user123:password1!@11.15.141.123:5432/gateway_db?schema=public&connection_limit=8
GATEWAY_URL	URL publik gateway geospa	https://10.14.151.119:3000 atau https://10.14.151.119/geo/api
AI_SERVICE_URL	URL publik layanan geospa AI	http://39.97.87.194:3008 atau http://39.97.87.194/geo/ai
VITE_GOOGLE_MAPS_API_KEY	Google Maps API Key sebagai penyedia map	AIzaSy-ContohAPIKeyGoogleMapsDev000
VITE_APP_BASE_URL	Path base rute akses frontend geospa. Diisi jika menggunakan path routing dengan nginx.	/geospatial/map

11.3.3. Dapatkan Client ID

Jalankan service backend dan gateway Geospa:

```
# Mengandung Formating '< >'(Harap diubah)!!!
cd /home/icon/<dir_name>
docker compose up -d gateway backend
```



Gambar 11.3. Service Backend & Gateway Berjalan

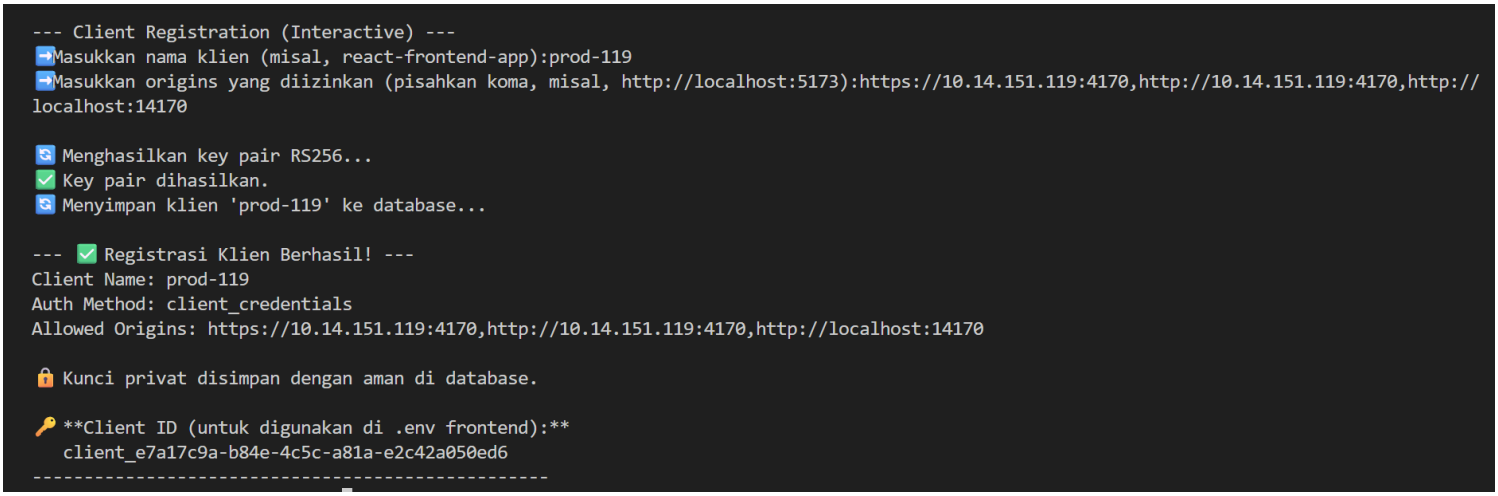
Migration database gateway:

```
docker compose exec gateway npx prisma db push
```

Registrasi client dengan perintah:

```
docker compose exec gateway npm run register_client
```

Masukkan nama klien: <nama client>
Masukkan origins yang diizinkan: <masukkan domain server, mis. https://10.14.151.119 >



Gambar 11.4. Registrasi Client

Salin dan simpan Client ID (dimulai dari client_...)

11.3.4. Konfigurasi .env Fase Kedua

Masukkan Client ID yang telah disalin dari Dapatkan Client ID
Edit dengan editor:

```
nano .env
```

Isi nilai variabel-variabel berikut:

Variabel	Deskripsi	Contoh

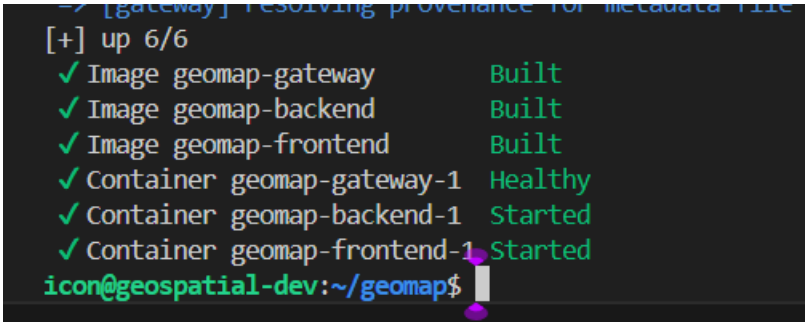
VITE_CLIENT_ID	Client id untuk authorization frontend	client_e7a17c9a-b84e-4c5c-a81a-e2c42a050ed6

Anda bisa melihat konfigurasi sebelum sesudahnya **DI SINI.**

12. Jalankan Deployment

12.1. Build & Up Aplikasi Geospa

```
docker compose up -d -build
```



Gambar 12.1. Build & Up Geospa

12.2. Publish Geospa ke Internet (Publik)

12.2.1. Salin Konfigurasi Routing Nginx Geospa ke Nginx Host

```
# Mengandung Formating '< >'(Harap diubah)!!!
sudo mkdir -p /etc/nginx/sites-available
sudo cp /home/icon/<dir_name>/nginx/geospa.conf /etc/nginx/sites-available/geospa
```

12.2.2. Aktivasi Konfigurasi Nginx Geospa

```
sudo mkdir -p /etc/nginx/sites-enabled
sudo ln -sf /etc/nginx/sites-available/geospa /etc/nginx/sites-enabled/
# Hapus konfigurasi default Ubuntu (agar tidak terjadi konflik port)
sudo rm -f /etc/nginx/sites-enabled/default
```

12.2.3. Verifikasi Konfigurasi

```
sudo nginx -t
```

12.2.4. Terapkan Konfigurasi Baru

```
sudo systemctl reload nginx
```

12.3. Verifikasi Service Geospa

```
docker compose ps
```

```
icon@geospatial-dev:~/geomap$ docker compose ps
NAME                IMAGE               COMMAND                  SERVICE    CREATED              STATUS                             PORTS
geomap-backend-1    geomap-backend      "docker-entrypoint.s…"   backend    59 seconds ago       Up 26 seconds (health: starting)   3001/tcp
geomap-frontend-1   geomap-frontend     "/docker-entrypoint.…"   frontend   58 seconds ago       Up 26 seconds (health: starting)   80/tcp, 127.0.0.1:14170->4170/tcp
geomap-gateway-1    geomap-gateway      "docker-entrypoint.s…"   gateway    About a minute ago   Up 57 seconds (healthy)            127.0.0.1:13000->3000/tcp
icon@geospatial-dev:~/geomap$
```

Gambar 12.2. Status Service Geospa

Akses:
Frontend: https://10.14.151.119:4170
Gateway: https://10.14.151.119:3000

13. Maintenance

13.1. Pengecekan & Pembaruan Manual SSL

```
# Mengandung Formating '< >'(Harap diubah)!!!
cd /home/icon/<dir_name>
./renew_ssl.sh
```

13.2. Pengecekan Status Service

13.2.1. Cek Service Nginx

```
sudo systemctl status nginx
```

13.2.2. Cek Service Docker

```
sudo systemctl status docker
```

13.2.3. Cek Service Geospa

```
# Mengandung Formating '< >'(Harap diubah)!!!
cd /home/icon/<dir_name>
docker compose ps
```

13.3. Monitoring Service Geospa

```
# Mengandung Formating '< >'(Harap diubah)!!!
cd /home/icon/<dir_name>
docker compose stats
```

13.4. Monitoring Logs Geospa

```
# Mengandung Formating '< >'(Harap diubah)!!!
cd /home/icon/<dir_name>
```

```
docker compose logs -f
```

```
# Jika ingin melihat layanan tertentu
# docker compose logs -f backend # gateway; frontend; backend
```

14. Troubleshooting

14.1. DNS Hijacking & Kegagalan Resolusi Nama

- **Masalah**: Muncul pesan Temporary failure in name resolution atau 404 Not Found saat menjalankan perintah apt update / apt install.
- **Penyebab**: Jaringan korporat atau ISP melakukan *DNS Hijacking* yang membelokkan permintaan DNS (seperti ke archive.ubuntu.com) menuju IP palsu.
- **Solusi**: Gunakan strategi *Static DNS*. Temukan IP mirror Ubuntu yang masih aktif (mengembalikan status HTTP 200) dengan mem-bypass DNS secara manual, lalu daftarkan IP tersebut langsung ke dalam file /etc/hosts server. Setelah didaftarkan, bersihkan *cache* paket menggunakan perintah sudo apt clean dan sudo rm -rf /var/lib/apt/lists/*, lalu jalankan kembali sudo apt update.

Jalankan perintah:

```
for ip in 185.125.190.36 185.125.190.39 91.189.91.81 91.189.91.82 91.189.91.83 91.189.91.84;
do
  echo -n "Testing $ip... "
  curl -s -o /dev/null -w "HTTP %{http_code}" --connect-timeout 5 \
    "http://$ip/ubuntu/dists/noble/InRelease" -H "Host: archive.ubuntu.com"
  echo ""
done
```

Daftarkan IP yang berhasil (status: 200):

```
echo "91.189.91.82 archive.ubuntu.com" | sudo tee -a /etc/hosts
echo "91.189.91.82 security.ubuntu.com" | sudo tee -a /etc/hosts
```

Hapus cache lama dan update repository:

```
# Hapus cache
sudo apt clean
sudo rm -rf /var/lib/apt/lists/*
```

```
# Update metadata dari NOL
sudo apt update
```

14.2. Akses Ditolak pada Docker (Permission Denied)

- **Masalah**: Terjadi *error* dengan pesan permission denied while trying to connect to the Docker daemon socket.
- **Penyebab**: Akun *user* yang Anda gunakan (contoh: icon) belum ditambahkan ke dalam grup keamanan docker.
- **Solusi**: Tambahkan *user* tersebut ke grup docker dengan menjalankan perintah sudo usermod -aG docker icon. **Sangat penting:** Anda wajib untuk *logout* dan *login* kembali ke dalam server agar perubahan grup tersebut diterapkan.

Jalankan perintah:

```
sudo usermod -aG docker icon
# logout dan login ulang
exit
```

14.3. APT Mengembalikan Status 404 Not Found (Repository Usang)

- **Masalah**: Instalasi gagal dengan *error* 404 Not Found ketika menggunakan apt install, sekalipun koneksi internet dan jaringan server repositori dalam keadaan normal.
- **Penyebab**: File metadata (cache) Ubuntu lokal Anda di /var/lib/apt/lists/ belum diperbarui dan masih mencari file .deb versi lama yang sudah tidak tersedia lagi di server pusat.
- **Solusi**: Hapus seluruh metadata lokal yang sudah usang dengan menjalankan sudo apt clean dan sudo rm -rf /var/lib/apt/lists/*. Segera setelah itu, perbarui ulang indeks repositori dengan menjalankan sudo apt update, kemudian ulangi instalasi paket Anda.

Jalankan perintah:

```
# Hapus semua cache metadata lama
sudo apt clean
sudo rm -rf /var/lib/apt/lists/*
# Download metadata segar dari repositori
sudo apt update
# Mengandung Formating '< >'(Harap diubah)!!!
# Ulangi instalasi (sekarang merujuk ke versi terbaru)
sudo apt install -y <package-name>
```

15. End of Document

If you encounter problem or difficulties, you may reach our support contact at :

1. WhatsApp or GSM contact to : +62 819 4401 1272
2. Email to support@kieroteknologi.com
3. Visit kieroteknologi.com and post your problem in contact us section